Problem 4

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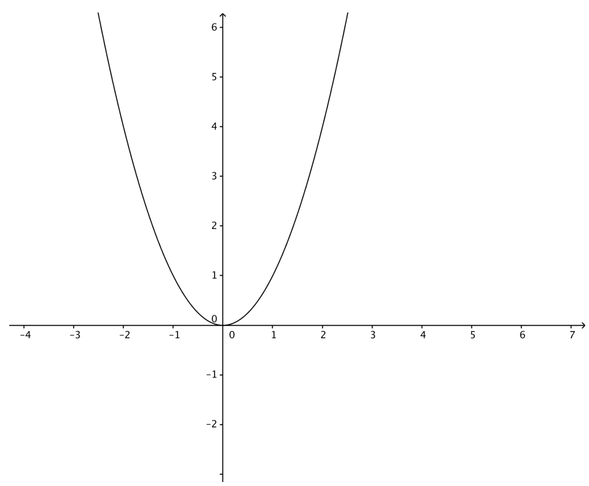
Problem 1 *The graph of a function f is the parabola given in the figure below.*

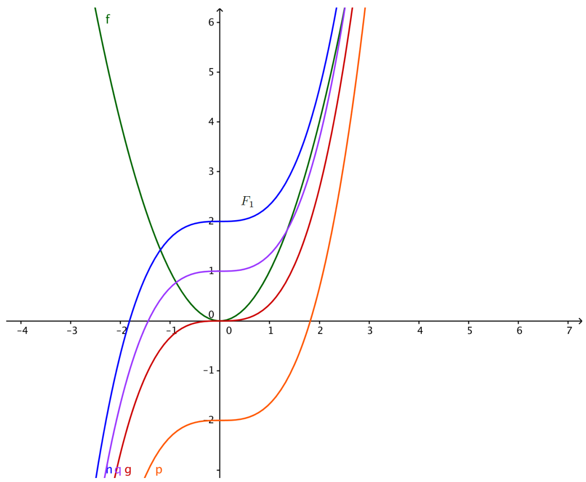


(a) *Find a formula for $f(x)$.*

Explanation. $f(x) = x^2$

(b) *Suppose F_1 is an anti-derivatives of f satisfying $F_1(0) = 2$. Sketch and label the graph of F_1.*
Sketch the graphs of three more antiderivatives of f.

Explanation.



(c) *Using the expression you found in part (a) and your sketches in part (b), find the algebraic representation of F_1.*

Explanation. The general antiderivative is $F(x) = \frac{1}{3}x^3 + C$. Therefore, $F_1(x) = \frac{1}{3}x^3 + C$, for some constant C.

Since $F_1(0) = 2$, it follows that $2 = \frac{1}{3}(0)^3 + C = C$. Therefore, $F_1(x) = \frac{1}{3}x^3 + 2$.

(d) *Suppose we're given $h(x) = (1/3)x^3 + 17$, what is $h'(x)$?*

Explanation. $h'(x) = x^2$.

(e) *What is the relationship between f, F_1, h, and h'?*

Explanation. We have $F_1' = f = h'$ and $h = F_1 + C$, where C is some constant. Determine this constant!